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SURVEY

Deep Learning and Transfer Learning in Cardiology: A Review of Cardiovascular Disease Prediction Models

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ABSTRACT Cardiovascular disorders are the primary cause of death on a global scale. The World Health Organization report indicates that approximately 18 million people die from CVD each year. Major cardiac risks include arrhythmia and coronary artery disease, among others. Recent advancements in Artificial Intelligence play a pivotal role for life-saving interventions in CVD treatment. This survey examines the latest progress in Machine Learning, Deep Learning and Pre-trained transfer learning models for classifying and predicting CVD using a survey of 159 articles, including 36 image datasets, 41 signal data and 52 clinical data from various sources. The survey investigates cardiac risk factors, cardiac dysfunction classification, various modalities and medical image processing techniques, performance metrics and hybrid techniques. Surveys on traditional neural networks such as Convolutional Neural Networks, Artificial Neural Networks, and Recurrent Neural Networks often achieve an accuracy rate of 70% to 95%. Leveraging pre-trained architectures such as ResNet, DenseNet, AlexNet, MobileNet, EfficientNet and GoogLeNet, transfer learning models consistently outperform other approaches frequently achieving accuracy levels greater than 96%. Researchers utilize various hybrid optimization algorithms to improve overall accuracy rate. The outcome of the survey supports a precise prognosis for patients with comorbidities. The survey findings indicate limitations in the incorporation of multimodal data for real-time risk assessment. The study results possess the capacity to bridge gaps in research on cardiovascular disease prediction, thereby assisting medical practitioners in early identification and subsequent prognosis.

INDEX TERMS Cardiovascular disease, machine learning, deep CNN, transfer learning, optimization hybrid model.

I. INTRODUCTION

Cardio Vascular Disease (CVD) are the primary cause of death across worldwide and poses an ineluctable challenge for medical sectors Sajid et al. [1]. WHO predicts CVD deathrate will hike around 24 million people by 2030. The heart has a crucial role in the human body as it pumps and distributes blood to all the other parts of the body. Irregular heart function includes a range of heart and blood vessel disorders Boulares et al. [2]. Approximately 85% of heart disease-related fatalities are caused by myocardial infarctions that lead to heart attack, stroke, and fainting affecting the cardiovascular system Abubaker and Babayigit [3]. Early detection of CVD has the potential to save many lives Dami and Yahaghizadeh [4]. A non-invasive technology at a lower cost is better than invasive techniques, of which are costly and uncomfortable Jain et al. [5]. Due to limited healthcare infrastructure, developing countries face a significant challenge to early diagnosis of CVD. Inadequate facilities, which include a lack of diagnostic equipment, insufficient staff, and geographical restrictions, make patients are unable to access essential healthcare treatments. Underdeveloped countries face higher cardiac risks due to malnutrition and contaminated water. Machine learning (ML) and Deep learning (DL)

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can investigate massive amount of patient's clinical data to provide personalized risk evaluation, optimize therapy plans, and assist in early diagnosis and prevention of cardiac risk.

Traditional methods of diagnosing cardiac disorders include Electrocardiogram, Echocardiogram, Carotid ultrasound, Coronary computed tomography angiography, Magnetic resonance imaging and several types of clinical tests. These approaches face challenges such as invasiveness, exposure to radiation, high costs, and inconsistencies in interpretation.

Traditional methods follow risk scores and methodologies that include Framingham Risk Score, American Heart Association Atherosclerotic Cardiovascular Disease (ASCVD) Risk Calculator, Registre Gironi del Cor, Systematic Coronary Risk Evaluation, and the United Kingdom Prospective Diabetes Study, which limits static nature, limited variables, assumption of linear relationships, risk of bias, lack of external validation, inconsistent reporting standards, limited use of advanced Data. To address the limitations of traditional risk prediction models, researchers are employing advanced machine, Deep and Transfer learning methods. These contemporary models examine large data sets to identify patterns, enhancing accuracy and precision in predicting cardiovascular risk.

Machine learning models are proposed for accurate classification of heart sounds. However, its effectiveness has been limited by the small dataset, which may impact its reliability in real-world clinical settings. The performance can vary across different datasets. This highlights the importance of considering the limitations of machine learning algorithms, such as dataset dependency and interpretability, when applying them to real-world clinical settings. Deep neural networks are proposed to learn complex feature structures from raw data, enhancing classification accuracy and the diagnosis of coronary artery disease (CAD) compared to traditional machine learning methods. Training deep learning models can be time-consuming, especially with large datasets and complex architectures. Deep learning models often require high-performance GPUs and substantial memory, along with the spatial and portability challenges of deep-learning computers, which can be considered limitations of deep learning. Transfer learning is a powerful method that can successfully address these kinds of challenges. Transfer learning involves applying a pretrained model, usually trained on an extensive dataset, and adapting it for a new, associated task. Transfer learning model attains great accuracy while significantly reducing computational resource requirements. Transfer learning model offers a lightweight and efficient solution, enabling accurate arrhythmia detection on mobile devices. This promotes timely heart health monitoring and early diagnosis.

This section provides a comprehensive overview of emerging trends in CVD prediction, and the primary research questions are addressed below.

Q1. What are the most prevalent identified risk factors for cardio vascular disorder prediction?

Q2. How can machine, deep learning and, Transfer learning techniques be applied to diagnose coronary artery disease (CAD) and peripheral arterial disease (PAD)?

Q3. What types of data modalities and sources have been applied to investigate the prediction of cardiovascular disease in the healthcare field?

Q4. How can we optimize the evaluation metrics to ensure the clinical reliability of ML, DL, and TL diagnostic methods?

Objectives

- Perform surveys on deep transfer learning methods for diagnosing CAD, PAD, and CVD risk factors.
- To investigate the effectiveness of medical image processing and software tools used to perform cardiovascular disease detection, with a specific focus on image segmentation, registration, and visualization techniques.
- To perform survey on effectiveness of ML, DL, and TL in predicting cardiac risks and improving the early detection of cardiovascular diseases.
- To compare the performance of ML, DL, and TL techniques in cardiac disease prediction and focusing on key performance metrics and their clinical implications.
- To survey the efficacy of hybrid deep learning models that incorporate optimization techniques for predicting cardiovascular illness.

A. CARDIOVASCULAR DISEASES: RISK FACTORS, DISORDERS AND DIAGNOSTIC APPROACHES

The dysfunctions of the heart and blood arteries are known as cardiovascular diseases

- Coronary Heart Disease caused by the accumulation of atheroma in the heart arteries.
- Cerebrovascular Disease affects the brain's blood vessels.
- Peripheral Arterial Disease impairs the arteries in arms, legs, and other limbs.
- Rheumatic Heart Disease damages heart valves and muscles due to rheumatic fever.
- Congenital Heart Disease ailment which affects function of the heart from birth.
- Deep Vein Thrombosis occurs formation of blood clots in deep veins.
- Pulmonary Embolism blood clots travel to the lungs and blocks pulmonary artery.

Fig. 1 provides various cardiovascular disorders, including coronary artery disease, deep vein thrombosis, peripheral arterial disease, rheumatic heart disease, cerebrovascular diseases, and congenital cardiac conditions.

B. CORONARY ARTERY DISEASE

Coronary artery disease (CAD), caused by plaque and triglyceride buildup in heart arteries, which restricts blood flow Trigka and Dritsas [6]. Chest discomfort is a common symptom of angina. While CAD often starts silently, stable angina occurs with exertion, and unstable angina can lead to heart

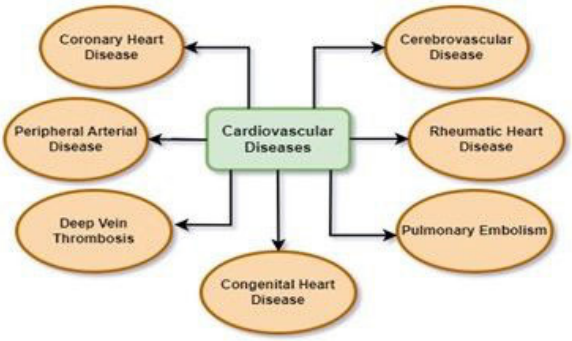


FIGURE 1. Cardiovascular disorders and their classifications.

attack Reeh et al. [7]. Acute myocardial infarction, caused by a blocked coronary artery, can result in sudden cardiac death Shao et al. [8]; Wong et al. [9].

C. PERIPHERAL ARTERIAL DISEASE

Peripheral artery disease (PAD) is caused by hardened arteries Al-Ramini et al. [10]. PAD is often underdiagnosed and undertreated, despite its risk for heart and brain problems Marso and Hiatt [11] PAD can be life-threatening, affecting blood flow to the head, neck, abdomen, and limbs Ara et al. [12].

D. RISK FACTORS FOR CARDIOVASCULAR DISEASE

1) HEART FAILURE

Heart failure occurs when the heart fails to pump sufficient blood, causing signs such as shortness of breath, fatigue, and swelling Ponikowski et al. [13]. Cardiologists classify heart failure into diastolic and systolic types. Diastolic heart failure is indeed linked to ventricular wall thickening, which limits the heart's diastolic function, and systolic cardiac failure is associated with ventricular dilatation and poor pumping function Genet et al. [14]. Heart failure symptoms vary depending on which side of the heart is affected. Left-side failure affects blood flow to the brain and body, while right-side failure affects blood flow to the lungs (Jessup and Brozena [15]; Lorell and Carabello [16]; Haddad et al. [17].

2) STROKE

A stroke is a severe brain ailment caused by interrupted blood flow to the Central Nervous System (CNS) Mir et al. [18]. Strokes can be categorized into two types, including Ischemic stroke and hemorrhagic stroke. Cerebral ischemia is caused by blocked blood arteries (Fan et al. [19]; Perna and Temple [20]. Hemorrhagic strokes cause brain irritation due to blood vessel bleeds Perna and Temple [20]. Transient Ischemic Attacks (TIA) is a brief neurological problem caused by temporary blood flow blockage to the brain Stanciu et al. [21].

3) HEART VALVE DISEASE

Heart valve disease can affect one or more of the heart's valves and disrupt blood flow by affecting the aortic, mitral, tricuspid, and pulmonary valves. Aortic valve is the last valve that oxygenated blood passes before leaving the heart. Mitral valve disease disrupts blood flow due to malfunctioning flaps between the left atrium, and tricuspid valve disease disrupts blood flow between the right atrium and ventricle Maeder et al. [22]; Turi [23]; Shah and Raney [24]. The pulmonary valve regulates blood circulation from the right heart to the lungs Naeije [25].

4) HYPERTENSION

Hypertension is a primary CVD risk factor that can cause damage to the heart, blood vessels, and organs. Primary hypertension is common, while secondary hypertension can be caused by underlying ailments, such as kidney disease and sleep apnea. White coat hypertension is induced by anxiety in the medical environment Schrader and Middeke [26]. Resistant hypertension is difficult to control with medication, even with three different types of drugs, including a diuretic Calhoun et al. [27]. Isolated systolic hypertension is prevalent among the elderly population Amery et al. [28]. Masked hypertension is detected by ambulatory blood pressure monitoring, which is conducted outside of the clinical setting Pickering et al. [29].

The article is organized into nine sections. Section II reviews existing research on CAD and PAD prediction, the latest imaging modalities, and classification techniques from various works in the literature. Section III focuses on cardiac risk prediction on medical images and analyzing software tools. Section IV discusses various utilized datasets and modeling techniques for cardiac risk prediction. Section V presents performance measures for assessing CVD risk factors. Section VI compares the performance of ML, DL, and TL applications. Section VII investigated an enhanced hybrid model using DL with a general architecture for CVD risk prediction. The conducted survey is detailed in section VIII, while further enhancements are concluded in section IX.

II. LITERATURE REVIEW

The literature review examined various cardiac dysfunction classifications for predicting cardiovascular disease.

A. CORONARY ARTERY DISEASE AND PERIPHERAL ARTERIAL DISEASE DIAGNOSIS USING DEEP LEARNING

We have surveyed relevant healthcare literature in Machine Learning (ML), Deep Learning (DL) techniques to understand current diagnostic approaches for cardiovascular disease such as angioplasty and Coronary Artery Bypass Grafting (CABG) plays a crucial role in reducing CVD occurrences LeCun et al. [30].Modern imaging procedures such as Echocardiogram, Magnetic Resonance Imaging, Computed Tomography, Phonocardiogram, cardiac catheterization and chatbot systems can facilitate the early diagnosis of CVD and support physicians in deciding how to overcome the critical diseases Boulares et al. [2]; Abubaker and Babayigit [3].

Arrhythmias, caused by abnormal electrical activity in the heart, can range from harmless to life-threatening. While most arrhythmias are not fatal, some, such as ventricular fibrillation and flutter, may cause sudden heart attack and hemodynamic collapse Go et al. [31]. CAD narrows the heart's arteries due to plaque accumulation, reducing blood flow to the heart muscle. Signs of CAD include an asymptomatic period with the presence of atherosclerotic lesions in the coronary arteries without symptoms Trigka and Dritsas [6]. Cardiologists classify heart failure into diastolic and systolic types Genet et al. [14]. Diastolic heart failure is linked to concentric hypertrophy, which impairs filling Grossman et al. [32]. Eccentric hypertrophy is linked to systolic cardiac failure and is seen as poor pump function Gerdes et al. [33]. A stroke is a severe CVD categorized as a catastrophic disease Mir et al. [18]. Stroke types include Ischemic Stroke, caused by cerebral blood artery thrombosis leads to brain ischemia, which is a major neurovascular cause of mortality and disability Fan et al. [19].

PAD is a form of CVD that occurs due to atherosclerosis, leading to reduced blood flow to the arteries and tissues Al-Ramini et al. [10]. The pulmonary valve functions as a one-way valve to allow blood to travel from the right chamber of the heart to the lungs for oxygenation. Hypertension is the primary cause of CVD and is classified into various types based on factors like cause, severity, and circumstances. Primary hypertension leads to serious harm to the heart, blood vessels, and other organs by adulthood, with changes noticeable even in children. Secondary hypertension in children may be caused by congenital abnormalities of the kidney, heart, and other organs, which should be thoroughly assessed Schrader and Middeke [26]. Table 1 summarizes different heart-related disorders.

III. CARDIOVASCULAR DISORDERS USING MEDICAL IMAGING

Analyzing medical images to predict cardiovascular disease holds significant importance in the healthcare field. This section investigates a hybrid deep learning approach to predict cardiovascular disease by leveraging medical image processing methods.

A. MEDICAL IMAGE PROCESSING FOR CVD PREDICTION

A Medical image processing system for the prediction of CVD is shown in Fig. 2. The process begins with the CVD image data source, which covers chest pain, obesity, fainting, shortness of breath, and fatigue. The CVD Scan report utilizes different imaging modalities, including MRI Soni et al. [46], PET Acharya et al. [47], along with cardiac stress tests. To diagnose CVD, different techniques are applied, including localization and segmentation. Localization identifies the area of interest in medical images, while segmentation classifies images into different sections Ari et al. [48]. Once segmented, features or irregularities within these segmented regions can be identified.

Feature extraction is a process that reduces data by converting original features into a smaller number of unique and useful features Dougherty [49]; Haq et al. [50]. Techniques such as relief, mRMR, LASSO and genetic algorithms can be used to select key features Masih et al. [51]. To enhance the data quality for early-stage coronary heart disease prediction, researchers have applied binning and z-score methods to address the inconsistencies of the acquired dataset.

Kalykakis et al. [153] explores to enhance the identification and characterization of significant coronary stenosis using Coronary Computed Tomography Angiography (CTCA). Hu et al. [154]. predicted a non-invasive technique for forecasting calcified plaques in coronary arteries using CT radiomics. The researchers analyzed radiomic features from perivascular fat CT images and integrated them with clinical risk factors to train and assess multiple machine learning models.

Kolossváry et al. [155] analyzed radiomic images to identify napkin-ring sign (NRS) plaques, which are key indicators of major adverse cardiac events. Through the analysis of quantitative features derived from CT images, major variations were identified between NRS and non-NRS plaques, proving the superiority of radiomics over traditional approaches. Militello et al. [156] examined the potential of radiomic features from pericoronaric adipose tissue surrounding the anterior interventricular artery (IVA) in evaluating coronary artery illnesses, in contrast to relying only on clinical risk factors.

B. MEDICAL IMAGING SOFTWARE TOOLS

CMRtools are used for ventricular function and myocardial perfusion analysis software produced by cardiovascular imaging solutions. Another tool for MRI is SuiteHEART, developed by NeoSoft.CVI42 produced by circle cardiovascular imaging supports for both MRI and CT. iNtuition tool developed by Terarecon that handles MRI, CT and SPECT. Segment software from Medviso is suitable for investigating MRI, CT, and SPECT data. Syngo.via, provided by Siemens, useful for MRI, CT, and SPECT modalities. IntelliSpace Portal from Philips works with MRI, CT and echocardiography data. VevoLAB from Visualsonics is useful for echo analysis. QLAB and TOMTEC, both produced by Philips, are also echocardiography tools Martin-Isla et al. [152].

The DL-based application can analyze and predict CVD using processed medical images and extracted features Acharya et al. [47]. The CNN model is used for ventricular arrhythmia classification and segmentation and provides information about shapes, volumes, and damaged regions of organs Tarroni et al. [52]. Localization and detection techniques are used to identify tiny lesions and localize specific objects within the medical image Ari et al. [48]. Arif et al. [53] used detection and localization techniques in myocardial infraction detection. The image registration technique is used to align object locations across multiple images into a logical and well-organized system Redlarski et al. [54].

TABLE 1. Cardiac dysfunction classification.

Author	Framework	Disease Classification	Data Source
Diker A et al. (2020) [34]	Extreme Learning Machine (ELM) with Differential Evolution for ECG Signal Classification	Arrhythmia	PTB Diagnostic ECG database
Hammad M et al. (2020) [35]	Multitier deep learning framework to identify arrhythmia	Arrhythmia	MIT-BIH
Hassan C A U et al. (2022) [36]	Enhance CVD prediction.- gradient boosted trees, MLP, RF	Coronary Artery Disease	UCI Repository
Caliskan A & Yuksel M E (2017) [37]	Develop a DNN classifier for CAD clinical records.	Coronary Artery Disease	Cleveland, Hungarian, Long Beach, and Switzerland
Nirschl J J et al. (2018) [38]	Creating a CNN classifier to identify heart failure	Left ventricular tissue (heart failure)	University of Pennsylvania
Choi, E et al. (2017) [39]	To forecast heart failure's (HF) early diagnosis	HF cases	Sutter Palo Alto Medical Foundation
Saragih G S et al. (2020) [40]	Neural network functionality of CNN is extended using random forest to classify IS data.	Stroke patients -CT Scan Images	RSCM hospital, Indonesia.
Choi J M et al. (2021) [41]	The ANN-based forecasting of HT post-AIS outperformed traditional ML techniques.	Stroke database	tertiary teaching hospital
Zhang D et al. (2022) [42]	To create and evaluate ML algorithms to find parameters that might customize PAD treatments and enhance outcomes.	PAD	National Inpatient Sample in US
Ara, L et al. (2019) [12]	ML and AI algorithms to predict PAD based on LEAD tests.	LEAD studies of PAD	Health electronic medical record system.
Roy T S et al. (2022) [43]	Improved CNN Xception model is used for efficient classification of various heart valve disorders.	Normal heart sound, Regurgitation in Mitral valve, bicuspid stenosis, Barlow syndrome, aortic stenosis specimen are in .wav format.	Github.
Fang M et al. (2023) [44]	To estimate hypertension risk for a person, factors including age and blood signs are used.	Hypertension data -EHR data	Huazhong University of Science and Technology, Union Shenzhen Hospital
Yun D et al. (2023) [45]	Comprehensible model helps doctors prepare for IDH and IDHTN.	Electrical medical record-EMR data	Seoul National University Hospital

IV. DATASET AND METHODS

Machine learning, Deep learning, and Transfer Learning (TL) models have shown significant potential in predicting cardiovascular risk. By analyzing vast datasets, these models can detect complex relationships, resulting in highly accurate and tailored predictive models for cardiovascular disease.

A. DATASET DESCRIPTION

Data quality and diversity significantly impact CVD detection. This section analyzes the efficiency of several deep learning approaches on different datasets.

Clinical datasets include Cleveland Haq et al. [50], Hungarian, Z-Alizadeh Sani, and Framingham (Rubini et al. [55]; El Hamdaoui et al. [56]), provide valuable clinical information for CVD diagnosis. Physiological signal datasets, including MIT-BIH Yıldırım et al. [57], PhysioNet PTB Safdarian et al. [58], and UCI repository Ramesh et al. [59], provide recordings of physiological signals like ECG and PCG essential for analyzing heart function. Acharya et al. [47] used 2-second ECG segments to differentiate shockable and non-shockable arrhythmias from CUDB, VFDB (both 250 Hz), and MITDB (360 Hz) databases and publicly available datasets accessible through platforms from Kaggle Lakshmanarao et al. [60] and IEEE data port. Sound signal analysis for heart disease prediction often used data from various sources E-scope and sample waveforms from PCG data Ari et al. [48]; Redlarski et al. [54].

B. MODELING TECHNIQUES

Preprocessing involves cleaning, modifying, combining, and minimizing original data to deal with issues like incomplete values, disturbances, and discrepancies. It plays a crucial role in creating an effective model for disease prognosis. Selecting relevant features is an essential part of the machine learning task as it enhances precision and decreases the computational resources needed for classification.

Machine learning models are chosen for their simplicity, interpretability, and effectiveness with structured data. Deep learning models, such as CNNs and RNNs, are effective in handling complex patterns in unstructured data but require significant computational resources and large datasets.

Transfer learning utilizes pre-trained models to improve performance and decrease training time, making it especially beneficial for scenarios with limited data. These models together provide enhanced performance, efficiency,

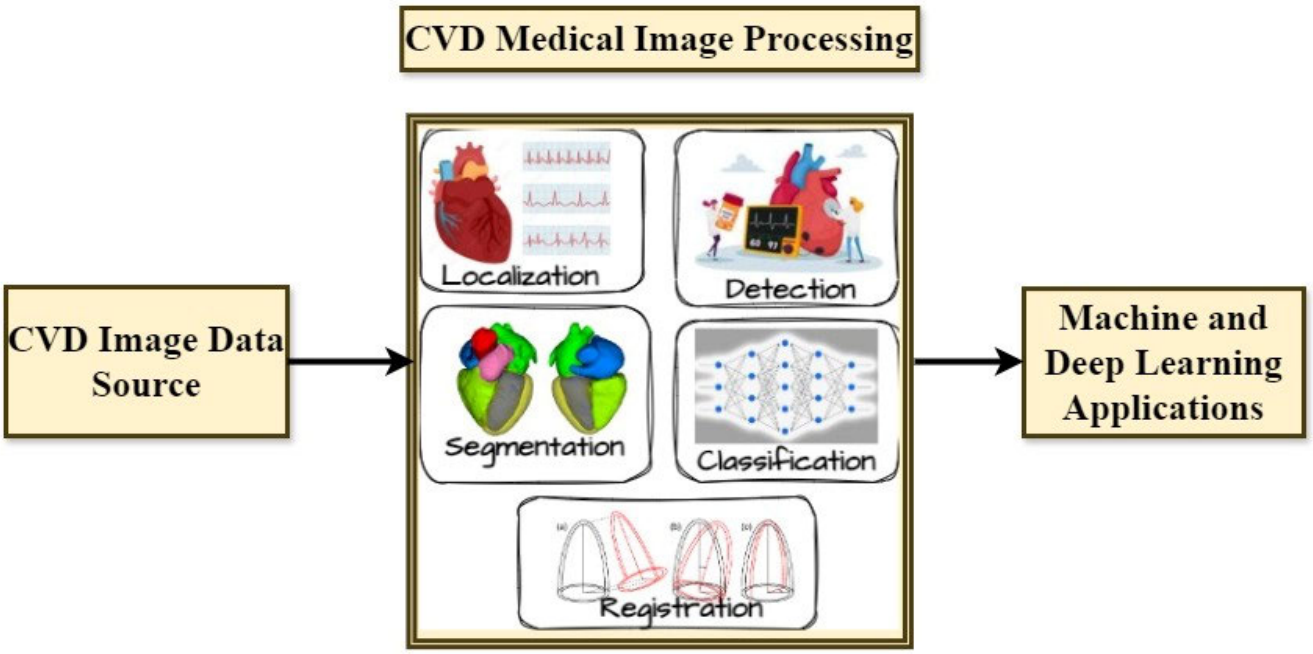


FIGURE 2. Evaluation of deep learning models in medical image processing for CVD prediction.

and application across diverse tasks in comparison to other techniques.

1) LOGISTIC REGRESSION

Logistic regression (LR) shown in (1), which predicts CVD risk using clinical data and patient features Hossen [61]. It estimates the probability of classifying input X into a particular group for binary and multi-class classification.

$$\mathrm{P}\,(\mathrm{x}) = \frac{\exp(\beta_0 + \beta_1 \mathrm{X})}{1 + \exp(\beta_0 + \beta_1 \mathrm{X})} \tag{1}$$

Here, β_0 represents bias and β_1 represents weight multiplied by input X.

2) DECISION TREE

The decision tree (DT) shown in Fig. 3 classifies patients into low, medium, and high-risk CVD groups using age, gender, and blood pressure as criteria Nikhar and Karandikar [62].

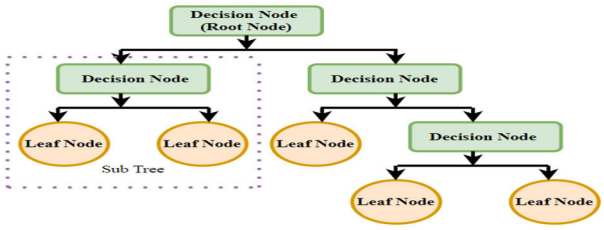


FIGURE 3. Decision tree for predicting risk of cardiovascular disease.

3) NAIVE BAYES

Naïve Bayes (NB) is a bayesian classifier shown in (2) that performs classification tasks, including electronic health information, and identifies diseases Lakshmanarao et al. [60].

$$P\,(c \mid x) = \frac{P\,(x \mid c)\,P(c)}{P(x)} \tag{2}$$

$P\,(x \mid c)$ – likelihood, P (c | x) – Posterior Probability, P(c) – class prior Probability, P(x) – predictor prior Probability.

4) ARTIFICIAL NEURAL NETWORKS

Artificial Neural Network (ANN) is often referred to as a neural network. The ANN model is shown in the Fig. 4, and (3) Mienye et al. [63] developed a sparse autoencoder-enhanced ANN based model used to diagnosis heart diseases.

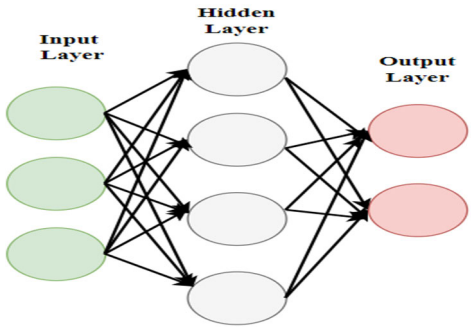


FIGURE 4. Artificial neural network model for predicting the risk of atherosclerotic cardiovascular disease.

Below function calculates output.

$$y_j \sum_{f=1}^{k} (W_{ij} X_i) \tag{3}$$

where,

y_j shows output neuron, X_i is input neuron, W_{ij} is the weight connecting X_i and y_j,

Σ is sigmoidal function

5) RANDOM FOREST

Random forest (RF) is used to predict CVD based on clinical data, as seen in Fig. 5. The final CVD prediction is calculated by averaging the probability from all decision trees in the ensemble model. Yadav and Pal [64].

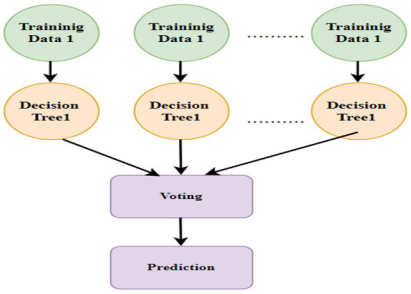


FIGURE 5. Random Forest model for cardiovascular disease risk prediction.

6) CONVOLUTIONAL NEURAL NETWORKS (CNN)

CNNs are becoming an effective technique for the early and precise detection of CVD by examining medical images like ECGs, echocardiograms, and cardiac MRIs Gu et al. [65]. Their functionality is represented in Fig. 6.

a: FUZZY CONVOLUTIONAL NEURAL NETWORKS

Wu et al. [157] The federated Epistasis Detection seg Neural Architecture Search (FedED-SegNAS) framework aims to enhance the accuracy and interpretability of epistasis detection in genomic data while preserving privacy. By integrating fuzzy logic into CNNs, the model can better handle uncertainty and complexity in genetic data, leading to a deeper understanding of the relationships between genetic variants of single nucleotide polymorphisms and complex diseases. The CVD ratio is high in females when compared to males as per genetic disorder.

7) K-NEAREST NEIGHBORS' (KNN)

The KNN approach for predicting cardiovascular disease based on clinical data involves calculating distance metrics between new patient data and all training cases, then identifying the K nearest training samples using Euclidean or Manhattan distances Heo, J et al. (2020) [66].

8) GRADIENT BOOSTING

Gradient Boosting (GB) addresses regression and classification problems by constructing an ensemble of decision trees that iteratively improve prediction accuracy by correcting errors from previous trees Feng et al. [121].

9) XGBOOST MODEL

XGBoost, as shown in (4), was applied to the CVD dataset to enhance prediction accuracy. Trees within this model identify patterns in clinical variables through iterative training, aiming to minimize the cross-validation error metric Bhatt et al. [68].

$$P(x) = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)} \tag{4}$$

10) GENETIC ALGORITHM

Genetic algorithms (GAs) utilize natural selection to develop predicative systems for cardiac risk. GAs iteratively improve models through processes of selection, crossover, and mutation, similar to how genes change over generations. By combining GAs with neural networks, an iterative approach can attain an accuracy rate of 93.8% in identifying coronary artery disease, as proved by Than et al. [67]; Arabasadi et al. [69].

11) MULTI-LAYER PERCEPTRON

MLP, a type of Artificial Neural Network (ANN), processes data in a feed-forward manner. Layers are interconnected, with hidden layers using non-linear activation functions to extract features from the data. Based on the learned patterns from the training data, the network classifies ECG data as either high or low risk for cardiac disease Kumari and Kumar [70].

12) TRANSFER LEARNING MODELS

Transfer learning is an effective method applied to several cardiovascular disease prediction systems, as listed in the Table 2. Researchers can improve the accuracy and efficiency of models for predicting and forecasting cardiac diseases by using knowledge derived from pre-trained models.

Predicting cardiac risk using neural networks has become a prominent method among researchers. Multi-Layer perceptions and ANN are fundamental neural network architectures that process data in feedforward manner, analyzing static data points such as blood pressure and cholesterol levels. Recurrent neural networks are particularly well suited for investigating time-series data in ECG readings. Many state-of-the-art deep learning approaches have emerged, providing enhanced feature extraction capabilities for heart risk, Yoo et al. [71].

ResNet prevents gradient deterioration by using skip connections. Jahmunah et al. [72]. DenseNet stimulates dense connectivity between layers, promoting information flow. Dhar et al. [73] AlexNet marked significant breakthroughs in image and PCG signal classification, particularly in cardiac images. De Marco et al. [74] MobileNet is a compact, efficient image classification model that uses average pooling and is compatible with mobile and wearable devices.

Wei et al. [75] The compound scaling approach of EfficientNet may enhance cardiac risk prediction by enhancing

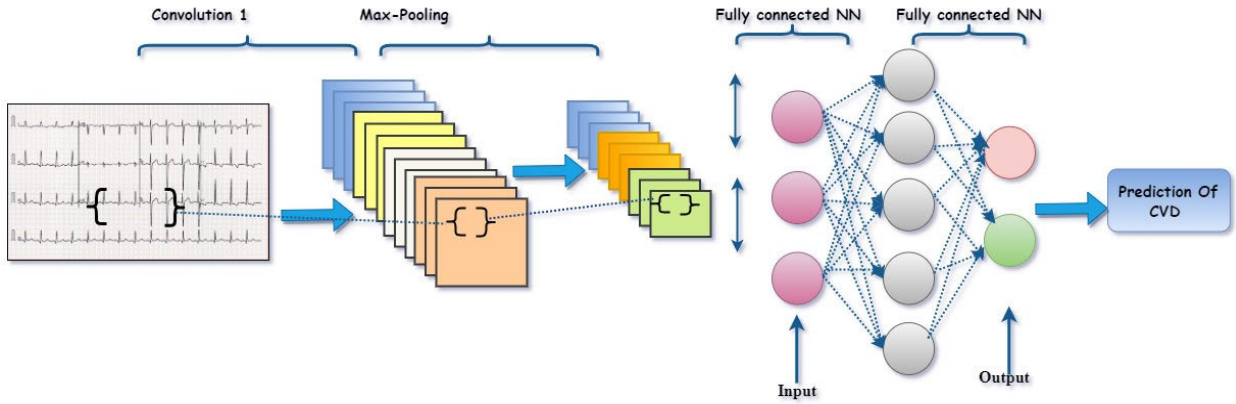


FIGURE 6. A fully connected CNN used for diagnosing cardiovascular disease.

network complexity. Dou et al. [76] The inclusion of inception modules in GoogLeNet makes it a comprehensive collection of neural network designs. Each architectural design has unique benefits for predicting cardiac disease, allowing researchers to investigate different methodologies and computing needs.

V. PERFORMANCE MEASURES

The assessment of deep learning models for cardiovascular prediction involves the use of performance metrics such as precision, accuracy, recall, specificity, and F1-score. These measures offer a comprehensive evaluation of the model's capacity to accurately categorize individuals who are at risk of cardiovascular disease. The models and metrics are discussed in this section. During training, achieving high accuracy and minimal loss indicates the model's ability to classify different stages of heart disease using samples from individuals with heart disease. The validation accuracy should closely approximate the training accuracy, which also avoids overfitting in CVD prediction. It analyzes in CVD prediction errors on the validation data. Which is desirable with minimal validity loss Abubaker and Babayigit [3].

A. CONFUSION MATRIX

The confusion matrix as shown in Fig. 7 provides an extensive evaluation of CVD prediction models. It visualizes actual versus predicted CVD status as positive or negative and helps in calculating accuracy, sensitivity, and specificity. El Hamdaoui et al. [58].

B. ACCURACY

The accuracy shown in (5) is to calculate the percentage of precise CVD risk predictions in the dataset. Abdar et al. [106].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{5}$$

where, TP-True positive; FP-False positive; TN-True negative; FN-False-negative.

C. PRECISION

Precision shown in (6) represents a fraction of positive predictions that indicate positive CVD risks. Abdulsalam et al. 2023 [92]

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

D. RECALL

Recall illustrated in (7) is sensitivity, which refers the value of the individual with positive CVD risk. Recall is termed to the True Positive rate (TPR). Abubaker and Babayigit [3].

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

E. F1-SCORE

The F1-score, as defined in (8), is a weighted average of both recall and precision for our CVD risk diagnosis Abdar et al. [106].

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

F. AUC

The Area under the Curve is a quantitative metric employed to assess the effectiveness of a binary classification model in predicting cardiovascular disease risk. Specifically, it measures the model's ability to differentiate between individuals with and without CVD risk by calculating the area under the ROC curve. This value ranges from 0 to 1. Zeleznik et al. [113].

G. ROC

The ROC curve illustrates the performance of the classification model in predicting CVD across various threshold values shown in (9) and (10). The graph depicts the correlation between the true positive rate and false positive rate across various threshold levels. true positive rate is also referred to

TABLE 2. Leveraging deep transfer learning to enhance CVD risk assessment via medical imaging.

Author	Dataset	Architecture
Abubaker M B et al. (2022) [3]	Public ECG images dataset	CNN with Squeeze Net, Alex Net-
Kiranyaz et al. (2015) [143]	MIT-BIH arrhythmia dataset.	CNN
Pal et al. (2021) [144]	MIT-BIH arrhythmia dataset.	Cardio Net with Dense net
Acharya et al. (2018) [47]	PTB dataset	CNN
Lessmann N et al. (2017) [145]	low-dose chest CT scans-National Lung Screening Trial	CNN
Wu D et al. (2019) [146]	436 CCTA scans	MLP with Tree Lab-Net and Bi-Tree LSTM
Mienye I D et al. (2020) [63]	CVD-Framingham, Massachusetts	SAE + ANN
Karimi-Bidhendi S et al. (2020) [147]	64 CMR studies from paediatric patients Hospital Los Angeles	DCGAN
Yang P et al. (2021) [114]	MIT-BIH	MKELM with RF
Kim J H et al. (2019) [148]	MIT-BIH	Google Net-DNN
Yoo, H et al. (2013) [71]	X-ray Image from National Institute of Health	Resnet with CNN
Zhang, Y et al. (2021) [149]	MIT/BIH arrhythmia	2-D ResNet101
Dhar P et al. (2021) [73]	PhysioNet	XWT + AlexNet
Jahmunah V et al. (2023) [72]	Physikalisch-Technische Bundesanstalt (PTB) database	DenseNet
Dou S et al. (2022) [76]	PhysioNet, MIT-BIH, BIDMC	CWT + GoogLeNet
Cui-fang Z et al. (2022) [150]	MIT-BIH	1D to 2D + AFF-EfficientNet
Wei, T. R et al. (2022) [75]	PhysioNet	EfficientNet B0
Shin S et al. (2022) [151]	MIT-BIH	MobileNetV2- BiLSTM
De Marco F et al. (2022) [74]	MIT-BIH	MobileNetv2

as sensitivity or recall:0 Shin et al. [151].

$$\text{TPR} = \frac{\text{True Positives} + \text{False Negatives}}{\text{True Positives}} \tag{9}$$

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}} \tag{10}$$

H. MEAN AVERAGE

The arithmetic mean is a statistical measure of central tendency obtained by adding all the values in a dataset and then dividing by the total number of values, as illustrated in (11). This measure is not utilized for predicting cardiac risk because it only represents the average rather than capturing variability in risk factors. Rajesh and Dhuli [140].

$$\bar{x} = \frac{\sum_{i-1}^{n} x^i}{n} \tag{11}$$

where,

x^i Each dataset value is represented.
$\sum_{i-1}^{n} x^i$ represents the dataset's total value.
n is the dataset's total value count

I. WEIGHTED AVERAGE

The weighted average shown in (12) is a modified form of the arithmetic mean in which each data point is allocated a specific weight based on its relative significance. Model prioritizes the most crucial factors for a personalized and accurate prediction of your CVD risk. Ramesh et al. [59].

$$\text{Weighted Average} = \frac{\sum_{i=1}^{n} w_i . x_i}{\sum_{i=1}^{n} w_i} \tag{12}$$

x_i Each value in the dataset is represented individually.
w_i Signifies the weight allocated to individual data points.
$\sum_{i=1}^{n} w_i$ represents the total of all weights

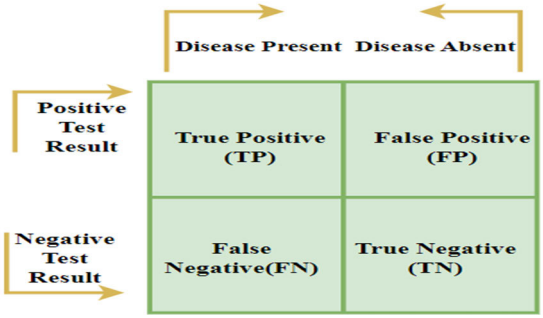


FIGURE 7. Confusion Matrix for predicting the risk of cardiovascular disease.

J. CORRELATION MATRIX

A correlation matrix (13) is shown in a square matrix that shows the correlation coefficients between variables in a dataset. It shows the relationship between each variable and all other variables. Hassan et al. [36].

The Pearson correlation coefficient formula calculates the correlation between A and B.

$$rAB = \frac{\sum_{i=1}^{n}(a_i - \bar{a})((b_i - \bar{b})}{\sqrt{\sum_{i=1}^{n}(a_i - \bar{a})^2 \sum_{i=1}^{n}(b_i - \bar{b})^2)}} \quad (13)$$

where,

a_i *and* b_i show specific values of variables A and B accordingly.

$\bar{a}$ and $\bar{b}$ shows the variables of means A and B.

n number of observations.

VI. PERFORMANCE COMPARISON OF DEEP LEARNING APPLICATIONS FOR CARDIOVASCULAR DISEASE PREDICTION

We have surveyed towards ML, DL, and TL techniques for predicting the CVD risk. ML algorithms are used to train for finding data patterns in prediction, classification, and dimensionality reduction LeCun et al. [30]. ML supports researchers to developing high-quality diagnostic tools for healthcare data analysis. The following contributions are examined by researchers using ML and DL techniques for CVD prediction. Kanagarathinam et al. [77] developed a CVD risk prediction model using a CatBoost ML classifier for CVD classification. Ordikhani et al. [78] implemented a genetic algorithm to predict CVD risk using the Explainable Persian Atherosclerotic CVD Risk Stratification based on the Isfahan Cohort Study of Iranian adults. Hasan and Bhattacharjee [79] proposed a method for classifying various CVD using a multi-dimensional deep CNN network that utilizes a revised ECG signal to predict CVD risk. Xia et al. [80] proposed as novel approach for Atrial Fibrillation (AF) detection. This technique utilizes the MIT-BIH with ECG intervals of 5 seconds, which also offers a rapid, accurate, and efficient tool for AF detection. Samuel et al. [81] developed a clinical decision support platform using a hybrid technique of Artificial Neural Networks and Fuzzy Analytic Hierarchy Process. Yildirim et al. [82] Proposed DNN model using 12-lead ECG signals to observe various rhythm classes from the cardiac arrhythmia. Paul and Karn [83] used scaled conjugate gradient backpropagation of the ANN approach to identify cardiac risk. Kuila et al. [84] detected cardiac arrhythmia using an ELM-CNN classifier of a 1D 12-layer neural network to minimize overfitting and improve the learning rate. Kong et al. [85] proposed temporal regression network architecture to detect End-Diastole and End-Systole frames from MRI sequences to monitor cardiovascular function. Vinay et al. [86] proposed supervised and unsupervised Recurrent Neural Network based Bidirectional Long Short-Term Memory machine learning algorithm, along with Generative Adversarial Networks for early and automatic detection of cardiovascular diseases using phonocardiogram signals, which are recordings of heart sounds. Chen et al. [87] developed a segmentation approach in neural networks to evaluate the structure and performance of the heart using cardiac MR images.

The K-means clustering algorithm, along with its variations such as Lloyd's approach, is used to eliminate dependencies and enhance the quality of clusters Zhang et al. [42]. Hinchcliff et al. [88] used unsupervised model-based clustering to investigate the role of the heart in systemic sclerosis, whereas Verma et al. [89] applied clustering techniques for cardiac disease. ECG image categorization and CVD detection have been investigated using quantum deep learning techniques such as Quanvolutional Neural Networks, quantum-behaved particle swarm optimization, and support vector machines Prabhu et al. [90]; Elsedimy et al. [91]; Abdulsalam et al. [92] developed an ensemble quantum ML approach to detect cardiac risk. To enhance the explainability of the model's results SHapley Additive exPlanations framework calculates and visualizes the importance of each parameter in prediction. Bhatt et al. [68] proposed a real-time CVD prediction system using Federated Learning (FL) and an ANN model, trained on real stroke case data. Yildirim et al. [82] proposed a 1D-CNN approach for fast arrhythmia classification using short ECG fragments (MIT-BIH data). Degirmenci et al. [93] developed a DL method using images derived from ECG signals to classify arrhythmias, including normal, PVC, LBBB, RBBB, and PB. Terrada et al. [94] investigated 835 clinical records of atherosclerosis patients from three data sources to design a medical diagnostic support system using ANN for improved CVD detection.

Tarroni et al. [52] developed a hybrid decision forest method to analyze short-axis cardiac MR image stacks. Dutta et al. [95] proposed CNN architecture for correctly classifying 77% of CHD patients and 81.8% of non-CHD cases in the testing data. Overall, the model achieves 85.7% accuracy in classifying the NHANES dataset to predict coronary heart disease. Ahmed et al. [96] used various techniques such as K nearest neighbour, support vector machine, multinomial naïve bayes, and gaussian naïve bayes attains maximum of 67.2% accuracy in CVD prediction. Ensemble models like Gradient Boosting Classifier (GBC) with AdaBoost Classifier (ABC) and Random Forest with GBC achieved nearly 87% accuracy, while CatBoost attains maximum of 87.93% accuracy. Chen et al. [97] performed cardiac MRI segmentation using CNN that attains a maximum accuracy using an image dataset. Blanchard et al. [98] diagnosed cardiovascular event risks during sleep apnea using the CNN model. The result shows well in AUC of 0.82 and sensitivity of 0.89. Ahmad et al. [99] predicted CVD using the hybrid CNN+BiLSTM technique with maximum accuracy.

Pan et al. [100] developed Enhanced Deep CNN for cloud-based heart disease diagnosis, achieving impressive performance metrics of 97.5% sensitivity, 93.5% accuracy, and 94.9% specificity. Ullah et al. [101] attained 97.3%-99.02% accuracy using 1D/2D CNNs for noisy ECG classification. Sarra et al. [102] developed an Enhanced-ANN model for cardiac detection attained 93.4% accuracy, 93.4% precision, 93.3% recall, 93% F1-score on Cleveland dataset. Clinical datasets such as the Cleveland, Hungarian, Z-Alizadeh

TABLE 3. Performance comparison for predicting cardiovascular disease.

Author	Objective	Dataset Used	ML and DL Techniques	Performance metrics	Limitations
Looker H C et al. (2015) [105]	To find predictive protein biomarkers for type 2 diabetes-related CVD.	Go-DARTS, SDR, MONICA/ KORA, IMPROVE, Stockholm	ML-elastic net	AUROC on fifty folds of test data	Limited to European cohorts, the study's generalizability to other populations is uncertain
Abdar, M et al. (2019) [106]	To identify CAD based on Iranian patient data	Z-Alizadeh Sani	SVM	Accuracy -93 %, F1 Score- 0.9151	To compare the suggested technique to decision trees (Random Forest, C4.5, C5.0) and MLP.
Kumar V D A et al (2022) [107]	To forecast the occurrence of early cardiac disease.	Heart Disease Training data	SVM	Accuracy -95%	Validate the algorithm in a huge population with multiple risk factors.
Pujadas E R et al. (2023) [108]	Applying CMR radiomics to predict various heart disease.	UK Biobank dataset., cardiovascular magnetic resonance (CMR) images.	SVM	AF and HF Accuracy- 0.71 and 0.79	CMR scans with clinical indications may enhance classification of risks in the future.
Anetta K et al. (2022) [109]	Predicting patient diagnoses from Polish electronic health records using automated ICD coding systems.	GCM- GH, Poland.	RoBERTa and BERT(NLP)	Accuracy - 78%	To enhance model accuracy, provide a collection of fine-grained diagnostic predictors targeting certain subsets.
Zhang Q et al. (2017) [110]	Novel multi-resolution convolutional neural network for biometric person identification.	CEBSDB, WECG, FANTASIA , NSRDB, STDB, MITDB, AFDB , VFDB	CNN	93.57% (Identification rate)	Enhanced user identification performance.
Zreik et al. (2018) [111]	Automatic identification of plaque in coronary artery and stenosis using RNN.	163 patients of CTCA scan data.	RCNN	Coronary artery plaque Accuracy- 0.77, Stenosis Accuracy - 0.80	Stenosis and coronary plaque are required for later cardiovascular examination.
Betancur J et al. (2018) [112]	To predict obstructive disease from myocardial perfusion imaging (MPI) to total perfusion deficit (TPD).	National and International (Canada, Switzerland, Israel) sites	CNN	Sensitivity improved for each patient from 78.8% to 83.3%	To test this strategy with different camera types and add clinical and imaging data.
Zeleznik R et al. (2021) [113]	To assess myocardial calcium on typical ECG gated CT and non-gated chest CT images	Framingham Heart Study	DCNN	AUC of 0.81-0.83	To improve patient care and outcomes by integrating into healthcare processes.
Yang H et al. (2021) [114]	Developed a biosensor with deep learning technique to identify atherosclerotic disease.	Cleveland dataset	ANN	Accuracy -95.6%	Different datasets and deep learning algorithms are used to test the proposed system output.
Lyu Q et al. (2021) [115]	RNN reduces motion artifacts and interpolates intermediate frames in cine cardiac MRI	ACDC dataset and Cedar's dataset	RNN	Mean + SD (Fine Detail) ACDC :4.62 +0.59 Cedar: 4.77 + 0.50	Design an efficient CMR imaging technique which improves image resolution.
Khan M U et al. (2022) [116]	Inexpensive self-developed Phonocardiography Analysis System	Real time PCG data.	ANN	Accuracy – 99.9%	To use reduction of features and fusion methods to improve accuracy performance
Esmaeili P et al. (2023) [117]	To predict ASCVD risk	Tabriz University Medical Sciences employees.	ANN	Accuracy – 98 %	Essential to change risk factors and enhance epidemiologic data quality.
Mishra I et al. (2023) [118]	To determine accurate stroke prediction method.	Odisha hospital data.	DT, SVM, AdaBoost, KNN, and NB	SVM Accuracy-96 %,	By comparing the k-modes clustering approach to other

TABLE 3. *(Continued.)* **Performance comparison for predicting cardiovascular disease.**

					popular clustering methods, the work may be improved.
Lui H W et al. (2018) [119]	Design a ML classifier for multiclass categorization of ECG recordings.	PTB database	CNN-LSTM stacking decoding	Sensitivity – 92 % Specificity-97.5% Precision-97 % F1 score-94.6%	Testing the techniques on a larger dataset would be beneficial
Rath A et al. (2021) [120]	To detect heat disease	MIT-BIH and PTB-ECG databases	GAN-LSTM	MIT-BIH: Accuracy-99.2% PTB-ECG: Accuracy: 99 %	Developed methods also applied for COVID 19, Pneumonia and other fatal diseases
Feng Y et al. (2022) [121]	Developing personalized survival prediction models for hypertension patients	Data from Alberta	LR, RF, GB, and ANN	RSF model: C-index > 0.70 and a Brier score < 0.25.	Study conducted only administrative data, which may not capture all relevant clinical information
Dami S et al. (2022) [4]	To predict cardiovascular events.	KAGGLE, Shahid Behesthi, Hospital, Physionet site, UCI	LSTM-DBN	Accuracy- 88.4%	Concentrate on collecting and analysing information of healthcare systems.
Srilakshmi V et al. (2022) [122]	An efficient model for predicting CVD risk using retinal fundus images. ,	JSIEC (1000 fundus images)	FC–HOA+DNFN	Accuracy – 91.6 %	Enhanced database for model efficiency evaluation.
Almulihi A et al (2022) [123]	Deep stacking ensemble model to improve heart disease prediction.	Dataset 1, Cleveland Dataset	hybrid models: CNN-LSTM and CNN-GRU	Dataset 1: Accuracy-79 %, . Cleveland: Accuracy-97 %,	Other datasets will be used to evaluate the suggested model.
Johri A M et al. (2022) [124]	To predict coronary artery CAD or CVD using DL-based algorithms (AE 3.0DL).	Multimodel data from Office, Laboratory and carotid ultrasound.	RNN, LSTM	RNN -Accuracy 95 % LSTM -Accuracy -95.3%	Validation and clinical implementation, exploration of new biomarkers
Li Q et al. (2022) [125]	To predict atherosclerotic CVD (apart from the coronary artery calcium (CAC) risk score.	Elizabeth Health Care System (Kentucky, USA)	LR, RF, GB, and NNs	AUC score of 0.902	To use unstructured data indications and symptoms to enhance our ML models' prediction capabilities.
Bhavekar G et al. (2022) [126]	Deep learning method is to effective heart disease prediction.	UCI Repository Heart Disease Data Set	RNN + LSTM	Accuracy -95 %	Analysing real-time IoT data for experiments will drive the system's future expansion.
Rai H M et al (2022) [127]	To identify heart malfunction in highly unbalanced data.	MITDB and PTBDB	CNN, CNN-LSTM, and ensemble approach	CNN Accuracy – 99 %	To generate minority class synthetic data samples to increase data samples and compare model performance across groups.
Oliveira B A et al. (2023) [128]	ML algorithms can automate CVD risk scoring for prevention.	Private company	LR, SVM, RF, XGB, Cat Boost and MLP	MLP: Accuracy- 80% F1-score of 70%	Assess the feasibility and acceptability of ML systems in healthcare.
Prabhu S et al. (2023) [90]	To classify ECG images to predict CVD	Omdena dataset	QPSO-SVM and QNN	QNN Accuracy -97.3%	Integrating quantum hardware and QML models into medical systems.
Elsedimy E I et al. (2023) [91]	To predict heart disease risk	Cleveland	QPSO-SVM	F1 Score -0.95	Using the existing frameworks to predict cardiac disease.

Sani Terrada et al. [94] and Framingham El Hamdaoui et al. [58] offer valuable clinical information for CVD diagnosis. Additionally, physiological signal datasets including MIT-BIH, PhysioNet PTB Safdarian et al. [58], Nagavelli et al. [103], and UCI repository Ramesh et al. [59], Subhadra and Vikas [104] provide recordings of physiological signals like

TABLE 4. Optimized hybrid models for CVD risk prediction.

Author	Dataset	Hybrid Techniques	Accuracy
Liu et al. (2015) [129]	Poly-ECG-S extracted features Akaike information criterion (AIC)	J48 DT	Accuracy- 94.4% Accuracy- 96 %
Lakshmanarao A et al. (2019) [60]	Kaggle	RF and DT	Accuracy- 91%
Safdarian N et al. (2014) [58]	PhysioNet PTB Database	NB	Accuracy- 95 %
Gonsalves A H et al. (2019) [130]	South African Heart Disease dataset-CHD	NB, SVM, DT	NB: Sensitivity - 63% Specificity - 76%
Ramesh TR et al. (2022) [59]	UCI	ANN	Accuracy- 86.5%
Abushariah M A et al. (2014) [131]	V.A. Medical Center, Long Beach and Cleveland	Neuro-Fuzzy system-ANN s	Training data: CC:100%(NFS) -90.7%(ANN) Test data: ACC:87.0(ANN) -75.9%(NFS)
Anbuselvan P et al. (2020) [132]	Cleveland dataset	RF	Accuracy- 86.8%
Rubini PE et al. (2021) [55]	Framingham	RF	Accuracy- 84.8 %
El Hamdaoui H et al. (2021) [56]	Cleveland and Stat log datasets	RF+AB	Accuracy- 96.6%
Yadav D C et al. (2020) [64]	UCI website	RF	Accuracy- 99.9%
Doppala B P et al. (2021) [133]	Cleveland	GA+RBF	Accuracy- 94%
Arabasadi Z et al. (2017) [69]	Z-Alizadeh Sani dataset.	NN+GA	Accuracy- 94%
Bhatt C M et al. (2023) [68]	Kaggle	XGB	Accuracy- 87%
Nagavelli U et al. (2022) [103]	PhysioNet database	XGB	Accuracy- 96%
Than M P et al. (2019) [67]	myocardial infarction	GB	AUROC:96.3
Arif M et al. (2012) [53]	PTB ECG database-MI	KNN	Sensitivity - 99.97% Specificity - 99.9%
Heo J Lee et al. (2020) [66]	PTB-diagnostic ECG	KNN	Accuracy- 96.3%
Wang H et al. (2020) [134]	MIT-BIH ECG database	CNN	Accuracy- 99 %
Acharya U R et al. (2017) [135]	MIT-BIH data source	CNN	Accuracy- 93 %
Hasan N I et al. (2019) [79]	MIT-BIH, PTB. St.-Petersburg	1D-CNN	Accuracy- MIT-BIH = 97 %, PTB = 98 %, St.-Petersburg = 99 %
Kumari V S R et al. (2013) [70]	MIT-BIH arrhythmia database	MLP	Accuracy- 95%
Verma L et al. (2016) [89]	Indira Gandhi Medical College, Shimla, India	CFS + PSO + Clustering + MLP	Accuracy- 90.8%
Subhadra K et al. (2019) [104]	UCI-Heart dataset	MLP+NN	Accuracy- 94%
Hossen M K et al. (2022) [61]	UCI	LR	Accuracy- 95%
Chauhan Y J et al. (2008) [136]	Framingham	LR	Accuracy- 89%
Amen K et al. (2020) [137]	Cleveland dataset	LR	Accuracy- 82
Sharma & Parmar M et al. (2020) [139]	UCI	Optimized DNN	Accuracy- 90.78%
Sun L et al. (2012) [139]	PTB diagnostic dataset	SVM	Sensitivity- 92 %, Specificity - 82 %
Rajesh, K. N., & Dhuli, R. (2018) [140]	MIT-BIH arrhythmia database	AB	Before / After Feature Selection: ACC:99 % / 98.6 %

ECG and PCG essential for analyzing heart function. Acharya et al. [47] used 2-second ECG intervals to differentiate shockable and non-shockable arrhythmias from CUDB, VFDB and MITDB databases. Lakshmanarao et al. [60] used Kaggle and IEEE DataPort with clinical and ImageNet datasets. Sound signal analysis, S1 and S2 of different pathological heart sounds used for CVD prediction Redlarski et al. [54]. Performance metrics of Machine Learning (ML) and Deep Learning (DL) models for CVD prediction are compared in this Table 3. It presents the benefits, limitations, and major discoveries of ML and DL techniques to predict CVD.

VII. A STUDY ON ENHANCED HYBRID MODELS

Cardiovascular disorders remain a leading global cause of mortality Oliveira et al. [128]. It is essential to create an enhanced hybrid deep learning model for precise prediction and early identification of cardiovascular disease Bhavekar and Goswami [126]. This model will merge the capabilities

of deep learning algorithms with other advanced methods to enhance precision and effectiveness in diagnosing cardiovascular disease. The enhanced hybrid deep learning model for cardiovascular disease will utilize convolutional neural networks Betancur et al. [112] and recurrent neural networks Lyu et al. [115] to extract pertinent features from medical data like electrocardiograms, echocardiograms, and patient characteristics. By integrating these diverse sources of data and utilizing cutting-edge deep learning techniques, the model will be capable of capturing intricate patterns and connections that indicate cardiovascular disease. Table 4 provides a summary of the hybrid models incorporating optimization.

Hybrid optimization approach for predicting CVD, as illustrated in Fig 8. The initial step involves acquiring data from clinical and image records, including ECG, PCG, MRI, CT, SPECT, DPCT, angiogram, and OCT data from various heart disease repositories. Data preprocessing involves removing of inconsistencies using outlier detection techniques such as Z-score, binning, and generative processing methods. For feature selection and extraction, the process begins by extracting features from the data, followed by selection of relevant features using techniques such as Minimum redundancy Maximum relevance (MrMr), relief, Particle Swarm Optimization (PSO) and Fast Correlation-Based Filtering (FCBF). Clustering K-means method clusters training instances for every class. A hybrid optimization approach combines bio-inspired algorithms such as the bat algorithm and spider monkey optimization to fine-tune the hyperparameters of hybrid deep learning models, leading to more precise CVD prediction. In the final step, the performance of the proposed models is evaluated using various metrics, including accuracy, precision, sensitivity, specificity, receiver operating characteristic (ROC) curve, and F1-score.

Metaheuristic algorithms such as Bat, Lion, Spider Monkey, Harris Hawks, Grey Wolf, Cuttlefish, and Bird swarm Optimization approaches are used to extract, select, and classify features to identify different types of CVD diseases. A few instances are identified in the works of Irshad et al. [141] improved the bat algorithm for cardiac beat feature extraction using microbat echolocation behaviour. Cuttlefish camouflage inspired the Cuttlefish Optimization Technique (COT), which improved COT for cardiac disease prediction with adaptive cross-layer stacked residual convolutional neural networks Srinivas et al. [142].

Wang and Wu [158] proposed the Ant Colony Optimization (ACO) algorithm is to improve the identification of Single Nucleotide Polymorphism (SNP) interactions in genomic data. This guarantees precise, effective, and privacy-conserving analysis. Genomic data can be used to predict CVD effectively.

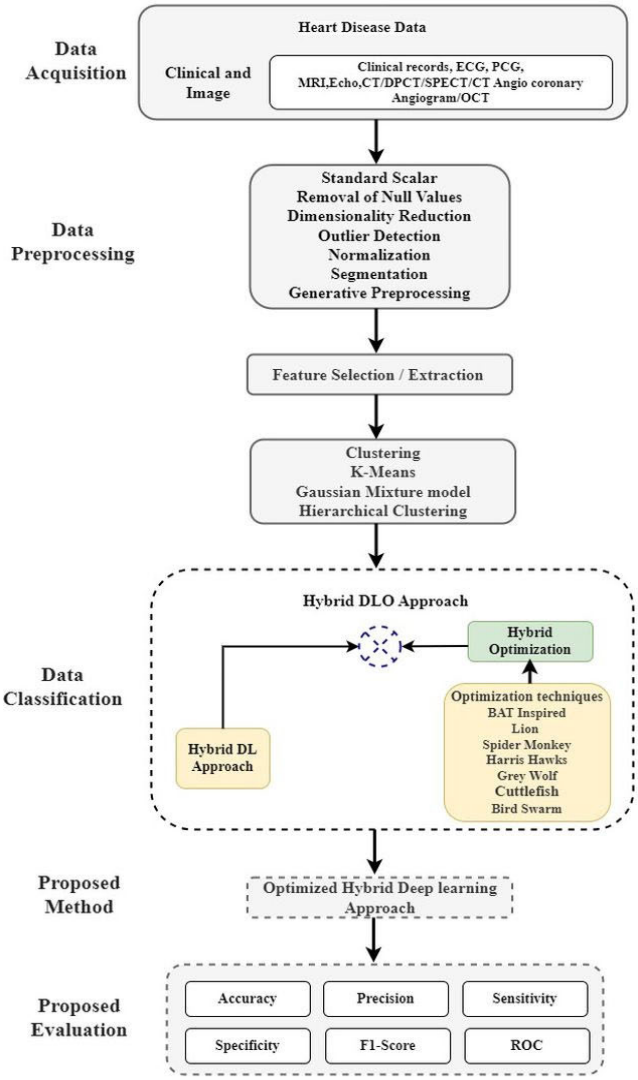


FIGURE 8. Hybrid optimized deep learning technique for predicting CVD.

VIII. DISCUSSION

This article conducts an extensive review of literature, encompassing prominent articles on ML, DL, and TL approaches for diagnosing cardiac diseases. The aim of the study was to categorize, evaluate, and explore various prognostic approaches utilized in CVD classification. The surveyed studies investigated various types of CVD, including CAD, PAD, and risk factors such as heart failure, stroke, heart valve disease, and blood pressure. The literature explored CAD and PAD disease prediction and classification of cardiac dysfunction. As per the survey blood pressure is the major parameter for cardiac dysfunction. Additionally, we examined cardiac disorders using medical imaging and software applications such as CMRtools, suiteHEART, CVI42 and iNtuition tools that are useful for MRI and CT imaging. VevoLAB, QLAB, TOMTEC are well suitable for echocardiography. Furthermore, iNtuition, Segment, Syngo. via are useful for SPECT analysis. From these options, CVI42 and iNtuition emerge as versatile solutions for managing diverse cardiovascular imaging data and analysis tasks. Our analysis involved a survey of diverse datasets, including clinical trials with patient demographics and disease characteristics data from Cleveland, Framingham, and UCI repositories. Signal data comprised of

ECG, PCG, and Heart rate data are obtained from MIT-BIH, PTB, while image data contained echocardiograms, CT, MRI, and coronary angiograms sourced from CADC and the UK Biobank. The researcher's performed localization, segmentation, detection, classification and registration on their works for cardiac medical imaging. A diverse range of modeling methodologies, including LR, DT, NB, ANN, RF, CNN, KNN, GB, XGBoost, GA, and MLP, was utilized to predict cardiac disorder. Among these, CNN models exhibited the best accuracy rate of greater than 95%.

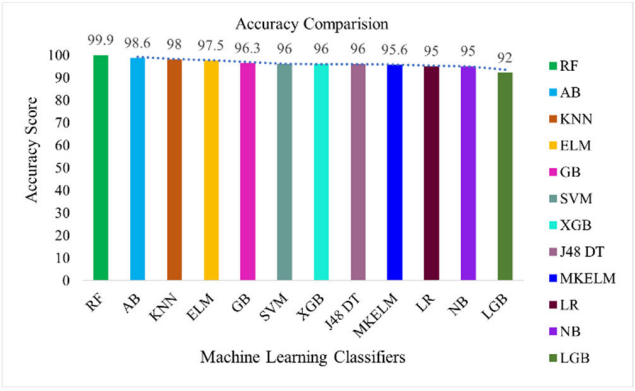


FIGURE 9. Accuracy of machine learning classifiers.

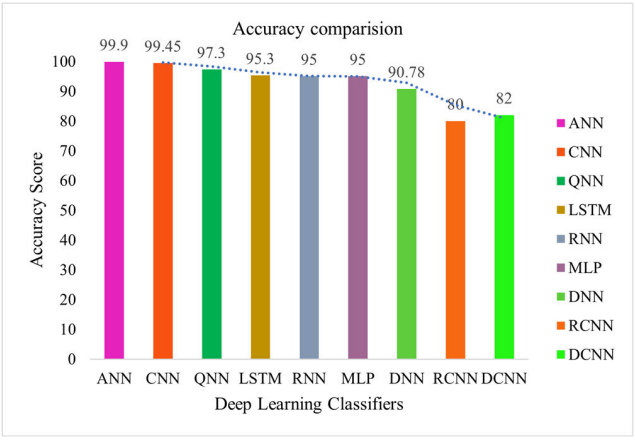


FIGURE 10. Accuracy of deep learning classifiers.

Furthermore, we explored the use of transfer learning models across multiple studies for cardiac risk assessment. The models such as EfficientNet, DenseNet, ResNet, AlexNet, GoogLeNet and MobileNet, from these models EfficientNet proves an accuracy rate of greater than 96 %. We surveyed the effectiveness of cardiac disease prediction using a standard set of metrics, including accuracy, precision, recall, F1-score, correlation matrix, area under the ROC curve (AUC), ROC, mean average, weighted average, and confusion matrix.

Finally, we analyzed advanced hybrid models such as RF+CNN, RF+AB, GA+RBF, NN+GA, MLP+NN, among these, RF+CNN achieves the accuracy rate greater than 100 %.

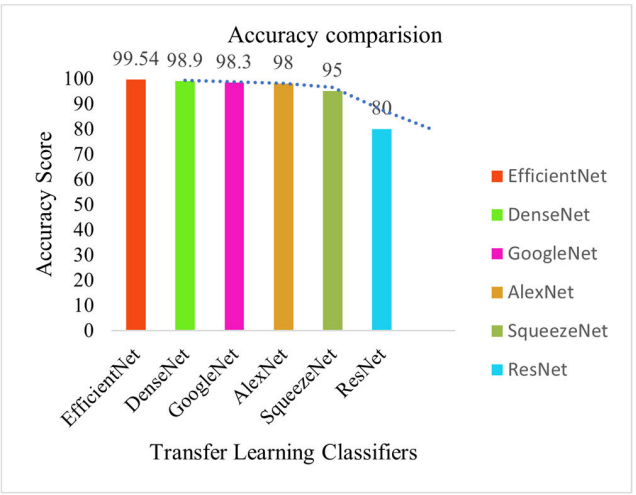


FIGURE 11. Accuracy of transfer learning classifiers.

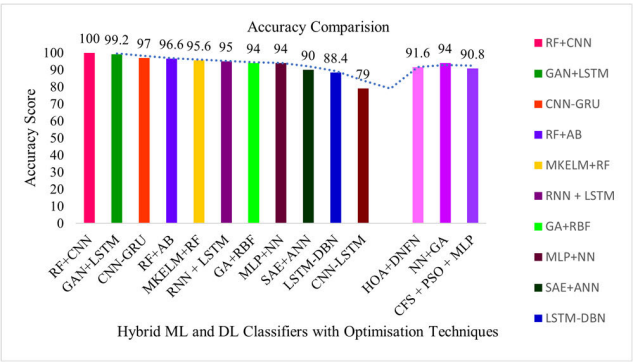


FIGURE 12. Hybrid ML and DL classifiers with optimization techniques.

The accuracy performance of various ML classifiers for cardiac illness detection is analyzed from Table 1, Table 2, Table 3 and Table 4. We observed that Random Forest, Adaboost, and K-Nearest Neighbor outperformed other ML classifiers, achieving accuracies of approximately 99.9%, 98.6%, and 98%, respectively, as illustrated in Fig. 9.

A comprehensive survey of Tables 1-4 reveals the performance of different DL classifiers in predicting CVD diagnosis. Artificial Neural Network and Convolutional Neural Network were the top performers, achieving accuracies of approximately 99.9% and 99.45%, respectively, as shown in Fig. 10.

EfficientNet, DenseNet, and GoogleNet were the best-performing TL classifiers in Table 2. Attained accuracies of approximately 99.54%, 98.9%, and 98.3%, respectively. These models demonstrated significant effectiveness for heart disease prediction, as illustrated in Fig. 11.

We analyzed all surveyed hybrid models that include machine learning, deep learning, and optimization approaches as shown in Fig. 12. The ML, DL and, TL models show an accuracy rate of 90 to 100. These findings highlight the efficacy of hybrid methodologies for enhanced performance.

TABLE 5. Comparative analysis of CVD classification using ML, DL, TL, and hybrid methodologies.

Disease/ AI Technique	HD	Arrhythmia	Myocardial Infraction	CAD	Stroke	PAD	CHD	Hypertension	Heart Failure
ML Classifier									
RF	✓	✓		✓	✓	✓			
AB	✓	✓			✓				
KNN		✓	✓					✓	
ELM	✓	✓						✓	
GB	✓		✓	✓					
SVM	✓		✓		✓		✓		
XGB	✓	✓							
DT	✓	✓			✓	✓	✓		
MKELM		✓							
LR	✓					✓		✓	
NB	✓	✓			✓		✓		
LGB						✓		✓	
DL Classifier									
DC+GAN	✓						✓		
ANN	✓	✓			✓			✓	
CNN		✓		✓					✓
QNN									
LSTM				✓					
RNN				✓					
MLP	✓	✓		✓					
DNN	✓			✓					
RCNN				✓					
DCNN	✓								
TL Classifier									
MobileNet		✓							
EfficientNet		✓							
DenseNet		✓							
GoogleNet		✓							
AlexNet		✓							
SqueezeNet		✓							
ResNet		✓		✓					
Hybrid ML and DL with Optimisation Technique									
RF+CNN					✓				
GAN+LSTM		✓							
CNN+GRU	✓								
RF+AB	✓								
MKELM+RF		✓							
RNN+LSTM	✓								

TABLE 5. *(Continued.)* **Comparative analysis of CVD classification using ML, DL, TL, and hybrid methodologies.**

GA+RBF	✓								
MLP+NN	✓								
SAE+ANN	✓								
LSTM+DBN	✓								
CNN+LSTM	✓	✓							
GRU+RNN									✓
GB+MLP				✓					
HNN+SNN						✓			
NN+GA	✓								
QPSO+SVM	✓								
HOA+DNFN	✓								
NN+GA	✓								
CFS + PSO + MLP	✓			✓					

The accuracy score of ML models varies between 92 to 99.9. DL models' accuracy score varies between 82 to 99.9. The accuracy score of TL models varies between 80 to 99.54. Hybrid models' accuracy score varies between 79 to 100. RF, ANN, and EfficientNet outperforms the other models.

The investigation shows that machine learning, deep learning, and hybrid optimization approaches have shown excellent efficacy in predicting cardiac disease. Random Forest and Artificial Neural Networks attained accuracies of around 99.9%, while the hybrid Random Forest and Convolutional Neural Network model enhanced the accuracy to 100%. These findings highlight the efficacy of hybrid approaches for precise and efficient cardiac disease prediction. The contemporary transfer learning models are more effective than other models in predicting CVD.

The comparative analysis for CVD classification is listed in Table 5.

IX. CONCLUSION

Cardiovascular disease poses a significant challenge for individuals and governing bodies striving to eradicate it. The survey analyzed the gaps and obstacles in predicting CVD. Accurate prediction of CVD is crucial for taking proactive measures. Several machine learning, deep learning, and transfer learning models have been examined, along with an exploration of optimization techniques. The survey compared the performance metrics of the existing approaches for CVD prediction. The survey indicates that ML and DL models achieve accuracy rates between 70% and 100%, where pre-trained transfer learning models show maximum validation accuracy (99.9%), sensitivity (96%), specificity (99%), F1-Score (90%), precision (95%), and

AUC (0.98). The study revealed that ML, DL, TL, and hybrid models integrating significantly enhance cardiovascular disease prediction accuracy. Since traditional models face many challenges, such as high cost, invasiveness.

Contemporary models provide a better solution in CVD predictions using clinical and imaging data. Deep transfer learning improves plaque identification and categorization, leading to better disease staging. The integration of AI and surgical robots will further assist traditional medicine.

Furthermore, the survey can be extended to analyzing deep hybrid models incorporating bio-inspired optimization techniques to enhance the prediction accuracy. Future research in cardiovascular disease prediction should focus on expanding dataset diversity to enhance model generalizability, and improving interpretability will assist physicians in understanding model decisions. Integrating real-time data from wearable devices may enhance the accuracy and promptness of cardiovascular disease forecasts. Explainable AI, attention mechanisms, transformer models, knowledge distillation, and digital twins are powerful AI techniques with significant potential in healthcare. Multimodal AI, integrating cardiac CT, cardiac magnetic resonance imaging (CMR), Genome-wide association studie (GWAS), and genomic AI, will further enhance the capabilities of AI in cardiac imaging, leading to more accurate diagnosis and personalized treatment plans.

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